

UNIVERSITÉ MARIE ET LOUIS PASTEUR

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Configuration Recognition Via Overlapping Boxes on Blinky Blocks in VisibleSim

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Abstract

This technical report details the design, implementation, and evaluation of a distributed configuration recognition algorithm tailored for 3D lattice-based modular robotic systems (Blinky Blocks) within the VisibleSim framework. Operating under decentralized neighborhood communication constraints where individual computational units lack global environmental awareness, the algorithm successfully computes a memory-efficient representation of complex structures as a compact set of overlapping rectangular boxes.

By executing asynchronous message passing across directly connected neighbors, modules collaboratively determine column heights, elect corner references, measure horizontal widths, and vertically stack layers along the $\vec{Z}$ axis. The system automatically processes internal holes and scales efficiently across multi-layered 3D volumes. Comprehensive simulation results across six distinct 2D and 3D configurations validate the precision, communication efficiency, and optimal parallel execution of the developed solution, while comparative analysis highlights its advantages over traditional border-tracing methods.

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1 Problem Statement and Objectives

Modular robotic systems frequently operate under decentralized conditions where individual computational units lack a global view of their environment.

- **The Problem:** How to compute and store a memory-efficient representation of a configuration's structure when each individual module possesses only a limited, neighbor-to-neighbor perspective, while correctly accounting for complex shapes containing both external perimeters and internal cavities (holes).
- **The Goal:** Design, implement, and evaluate a distributed message-passing algorithm where modular robotic units collaborate asynchronously to describe the structure as a compact set of overlapping boxes in 2D and 3D without any central oversight.

2 System Assumptions

The implementation relies on several core architectural and operational assumptions regarding the modular robot platform:

- **Lattice Topology:** Modules are positioned on the cells of a regular grid, where each module possesses precise local knowledge of its coordinates and orientation.
- **Localized Communication:** Communication is strictly restricted to neighbor-to-neighbor message passing via physical connectors, allowing receiving nodes to identify relative directions.
- **Decentralized Processing:** All computations are performed locally based solely on information obtained through local neighborhood interactions.
- **Static Configuration:** The network topology remains fixed and fully connected throughout the execution of the algorithm.

3 Algorithm Description and Mathematical Formalism

The main objective of the algorithm is to identify a set of full volumetric boxes that completely cover the configuration. Let M denote the set of modules sharing a 3D coordinate system. A Box B is formally defined by two boundary vectors $C_{min}(x_{min}, y_{min}, z_{min})$ and $C_{max}(x_{max}, y_{max}, z_{max})$, such that any module coordinate $X(x, y, z) \in B$ satisfies:

$$X(x,y,z) \in B \iff \begin{cases} x_{min} \leq x \leq x_{max} \\ y_{min} \leq y \leq y_{max} \\ z_{min} \leq z \leq z_{max} \end{cases}$$

3.1 Column Height Calculation (d_m)

The initial step determines the rank d_m for every module $m \in M$, representing its vertical distance moving backward along the $\vec{Y}$ axis until an empty position is reached:

1. Modules without an attached neighbor in the backward direction set $d = 1$.
2. They transmit a message SET_D_MSG(1) to their front neighbor. Upon receipt, subsequent modules set $d = d_{sent} + 1$ and forward the message until encountering an empty cell.

3.2 Corner Election (R_{min}) and Width Measurement (w_m)

Once d_m is established, a module m locally determines if it is a corner (R_{min}) based on its front neighbor and left neighbor height d_{left}:

$$\text{empty(front)} \wedge (d \neq d_{left} \vee \neg\text{empty(left + front)})$$

Corner modules initiate a FIND_W_MSG(id, d) rightward to compute the maximum horizontal span w_m of connected columns having equal or greater height:

$$w_m = \max_{n \text{ connected}} \Big(y_m = y_n \wedge (d_i \geq d_m \forall i \in [m, n]) \Big)$$

3.3 Vertical Layer Stacking in 3D

To extend the 2D rectangular decomposition into 3D, modules communicate upward along the $\vec{Z}$ axis. If adjacent layers share identical bounding rectangles, they merge into a single taller box of height h, yielding the final volumetric box definition:

$$\text{myBox} = \big(\{x, y, z\}, \{x + w - 1, y + d - 1, z + h - 1\}\big)$$

4 Validation and Simulation Results

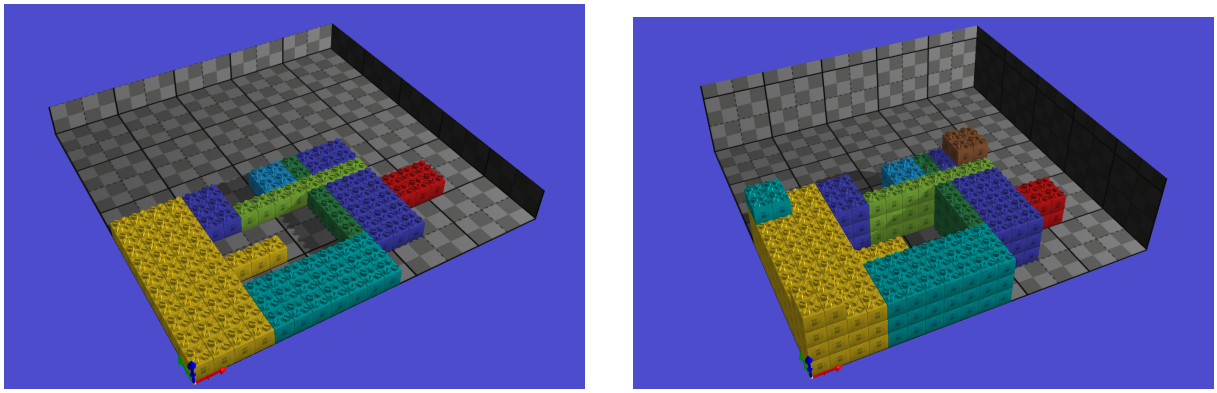
The implementation was rigorously tested within the VisibleSim framework using Blinky Blocks modules across six distinct configurations combining 2D planar layouts and 3D volumetric extrusions.

Structure	Description	Robots	Boxes	Messages	Time
config_1	2D shape with several holes	115	14	636	47.5 ms
config_1_3d	3D volumetric extrusion of config_1	348	17	2705	61.1 ms
config_2	Complex 2D maze layout	718	28	4550	211.5 ms
config_2_3d	3D extruded maze structure	2154	34	19382	222.2 ms
config_3	2D perimeter enclosure	532	22	3574	157.9 ms
config_3_3d	3D castle structure with towers	2764	42	23987	174.5 ms

Table 1: Results measured in VisibleSim. Time represents the simulated execution duration; message and box counts are exact and identical across runs.

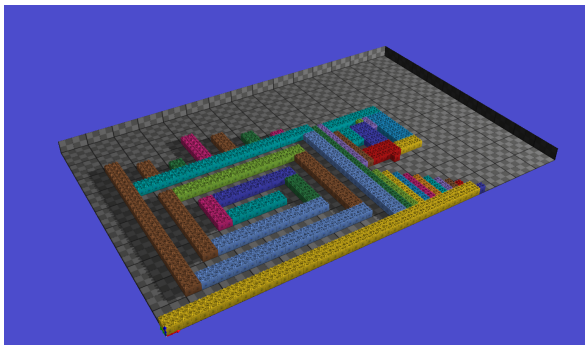
5 Visual Comparison of 2D and 3D Configurations

The visual outputs captured from VisibleSim illustrate how the distributed algorithm segments different shapes into colored bounding boxes across 2D planes and 3D volumes.

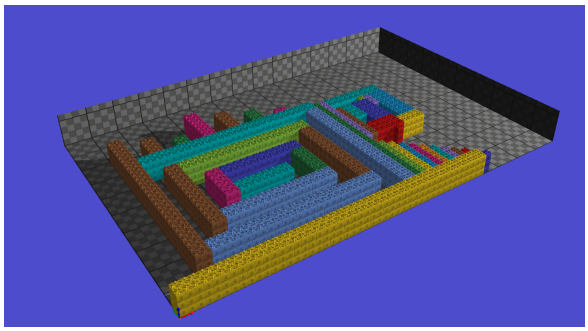


(a) Configuration 1 (2D) (b) Configuration 1 (3D)

Figure 1: Comparison of 2D shape decomposition versus 3D volumetric box stacking for Test Case 1.

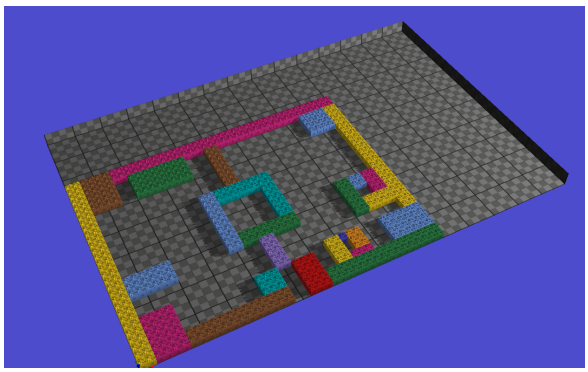


(a) Configuration 2 (2D)

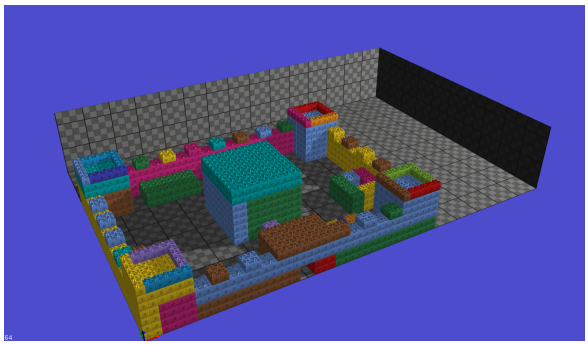


(b) Configuration 3 (3D)

Figure 2: Comparison of complex maze layouts across 2D slices and extruded 3D layers.



(a) Configuration 3 (2D)



(b) Configuration 3 (3D)

Figure 3: Comparison of castle configurations showing vertical tower and wall extensions.

6 Comparison with Border Tracing

- **Idea:** Border tracing sends a single message walking around the outline perimeter, whereas overlapping boxes have every robot measure its local column, row, and stack simultaneously in parallel.
- **Time:** Border tracing is slow because it processes modules sequentially along the outline ($O(P)$), while overlapping boxes execute rapidly in parallel ($O(L_x + L_y + H)$).
- **Memory:** Both maintain small per-robot memory, but border tracing messages grow with the number of outline segments while overlapping boxes keep a fixed $O(1)$ footprint.
- **Holes and 3D:** Border tracing requires a separate walk for every internal hole and struggles with 3D extensions. Overlapping boxes handle internal holes automatically and stack vertically into 3D volumes naturally.

7 Conclusion

This project successfully demonstrated the feasibility and robustness of decentralized configuration recognition in modular robotic systems using overlapping boxes. By implementing neighbor-to-neighbor message passing, Blinky Blocks successfully coordinated to identify complex structures, process internal holes automatically, and scale efficiently to 3D volumetric shapes without centralized oversight. Future work will investigate dynamic reconfiguration and fault-tolerant message handling.